

CIL MT 2026 — TOPIC-WISE PRACTICE SHEETS | PAPER-II (System / CS-IT)

Coal India Limited | Management Trainee | System Discipline (Post Code 14) | Advt. No. 03/2026

One sheet per topic, in syllabus order. Read the theory and formulas for a topic first, then attempt that sheet in one go — the whole sheet is the same topic, so a wrong answer points straight at the concept to revisit.

Total questions in this booklet: 300 across 10 topic sheets. Each question carries 1 mark in the exam and there is **no negative marking**, so attempt every question.

Suggested pace: about 50 seconds per question. The target time for each sheet is printed in its heading.

All answer keys are in one section at the end, listed sheet by sheet.

SHEET 1 — ENGINEERING MATHEMATICS

Professional Knowledge — System (CS/IT) | 36 questions | target time 30 minutes

- Q1.** The number of distinct arrangements of the letters of the word "LEVEL" is
(A) 60 (B) 20 (C) 30 (D) 120
- Q2.** For events A and B with $P(B) > 0$, the conditional probability $P(A | B)$ equals
(A) $(B) P(A \cap B)/P(A)$ (B) $P(A) \cdot P(B)$ (C) $P(A \cap B)/P(B)$ (D) $P(A \cup B)/P$
- Q3.** Near a simple root, the Newton-Raphson method converges
(A) linearly (B) cubically (C) quadratically (D) not at all
- Q4.** The number of distinct labelled spanning trees of the complete graph K_4 is
(A) 64 (B) 16 (C) 4 (D) 8
- Q5.** The eigenvalues of the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ are
(A) 1 and 3 (B) 1 and 2 (C) 2 and 3 (D) 0 and 3
- Q6.** A tree has 10 edges. The number of vertices in the tree is
(A) 11 (B) 12 (C) 10 (D) 9
- Q7.** The determinant of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is
(A) 0 (B) 24 (C) -6 (D) 1
- Q8.** For a symmetric matrix A
(A) A is always orthogonal (B) $A^T = -A$ (C) $A^T = A$ (D) A is always singular
- Q9.** The number of reflexive relations that can be defined on a set of 4 elements is
(A) 2^{16} (B) 2^4 (C) 2^{10} (D) 2^{12}
- Q10.** The sum of all eigenvalues of a square matrix is equal to its
(A) trace (B) determinant (C) norm (D) rank
- Q11.** The standard deviation of the data 2, 4, 4, 4, 5, 5, 7, 9 is
(A) 1 (B) 4 (C) 2.5 (D) 2
- Q12.** Simpson's $1/3$ rule for numerical integration requires the number of subintervals to be
(A) odd (B) prime (C) even (D) any positive integer
- Q13.** The number of elements in the power set of a set with 5 elements is
(A) 64 (B) 10 (C) 32 (D) 25
- Q14.** A square matrix A is singular if
(A) $\det(A) \neq 0$ (B) A is orthogonal (C) A is symmetric (D) $\det(A) = 0$
- Q15.** The rank of the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is
(A) 0 (B) 2 (C) not defined (D) 1
- Q16.** A and B are independent events with $P(A) = 0.4$ and $P(B) = 0.5$. Then $P(A \cup B)$ is
(A) 0.90 (B) 0.70 (C) 0.60 (D) 0.20
- Q17.** The variance of a random variable that takes a single constant value with probability 1 is
(A) undefined (B) 1 (C) 0 (D) equal to the mean
- Q18.** If A is a 3×3 matrix with $\det(A) = 2$, then $\det(2A)$ is
(A) 16 (B) 32 (C) 8 (D) 4
- Q19.** The value of the definite integral $\int_0^1 x \, dx$ is
(A) $1/2$ (B) $1/3$ (C) 2 (D) 1
- Q20.** If $|A| = 3$ and $|B| = 4$, the number of functions from A to B is
(A) 24 (B) 12 (C) 81 (D) 64
- Q21.** For the recurrence $a_n = a_{n-1} + 2$ with $a_1 = 1$, the value of a_{10} is
(A) 21 (B) 17 (C) 20 (D) 19
- Q22.** A bag contains 3 red and 2 blue balls. Two balls are drawn one after another without replacement. The probability that both are red is
(A) $9/25$ (B) $3/10$ (C) $1/2$ (D) $2/5$
- Q23.** In a partially ordered set, the least upper bound of a pair of elements is called the
(A) infimum (B) complement (C) minimal element (D) supremum
- Q24.** For a Poisson distribution
(A) variance = mean² (B) mean < variance (C) mean = variance (D) mean > variance
- Q25.** The number of ways of selecting 3 objects out of 8 distinct objects is
(A) 112 (B) 336 (C) 24 (D) 56
- Q26.** The eigenvalues of the matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ are
(A) 1 and 3 (B) 1 and 2 (C) 0 and 4 (D) 2 and 2
- Q27.** The total number of binary relations that can be defined on a set of 3 elements is
(A) 512 (B) 9 (C) 64 (D) 27
- Q28.** The number of edges in the complete graph K_7 is
(A) 14 (B) 21 (C) 42 (D) 49

Q29. A fair coin is tossed 4 times. The probability of getting exactly 2 heads is

(A) $1/2$ (B) $5/16$ (C) $1/4$ (D) $3/8$

Q30. The relation “congruent modulo 5” defined on the set of integers is

(A) a partial order only (B) antisymmetric but not reflexive (C) not transitive (D) an equivalence relation

Q31. The number of ways in which 5 distinct books can be arranged on a shelf is

(A) 120 (B) 720 (C) 25 (D) 60

Q32. The number of edges in the complete bipartite graph $K_{3,3}$ is

(A) 9 (B) 12 (C) 18 (D) 6

Q33. For the system $AX = B$ where A is square with $\det(A) = 0$ and $B \neq 0$, the system has

(A) a unique solution (B) always infinitely many solutions (C) exactly two solutions (D) either no solution or infinitely many solutions

Q34. The number of onto (surjective) functions from a set of 4 elements to a set of 3 elements is

(A) 36 (B) 64 (C) 24 (D) 81

Q35. For a binomial distribution with $n = 10$ and $p = 0.4$, the mean is

(A) 2.4 (B) 10 (C) 0.4 (D) 4

Q36. Two fair dice are rolled. The probability that the sum of the numbers is 7 is

(A) $1/12$ (B) $1/9$ (C) $5/36$ (D) $1/6$

SHEET 2 — DIGITAL LOGIC

Professional Knowledge — System (CS/IT) | 24 questions | target time 20 minutes

- Q1.** The Boolean expression $A'B + AB$ simplifies to
(A) AB (B) B (C) A (D) $A + B$
- Q2.** The number of 2-input NAND gates required to realize a NOT gate is
(A) 3 (B) 4 (C) 2 (D) 1
- Q3.** In the IEEE 754 single-precision format, the number of bits allotted to the exponent is
(A) 8 (B) 7 (C) 11 (D) 23
- Q4.** The number of distinct states in a MOD-16 counter is
(A) 15 (B) 4 (C) 8 (D) 16
- Q5.** The dual of the Boolean expression $A + (B \cdot C)$ is
(A) $A \cdot (B + C)$ (B) $A \cdot (B \cdot C)$ (C) $A + (B + C)$ (D) $A' + (B' \cdot C')$
- Q6.** The range of integers representable in 8-bit 2's complement form is
(A) 0 to 255 (B) -127 to $+127$ (C) -128 to $+127$ (D) -128 to $+128$
- Q7.** The hexadecimal equivalent of the binary number 11011010 is
(A) AD (B) DA (C) D8 (D) BD
- Q8.** A full adder can be realized using
(A) two half adders and one OR gate (B) one half adder and one AND gate (C) two half adders only (D) three half adders
- Q9.** The 1's complement of the binary number 0101 is
(A) 1010 (B) 1011 (C) 1101 (D) 0110
- Q10.** The number of select lines required for an 8-to-1 multiplexer is
(A) 2 (B) 4 (C) 8 (D) 3
- Q11.** The number of flip-flops required to build a 4-bit shift register is
(A) 16 (B) 2 (C) 4 (D) 8
- Q12.** The output of a 2-input XNOR gate is 1 when
(A) both inputs are 0 only (B) the inputs differ (C) both inputs are 1 only (D) the inputs are equal
- Q13.** The Gray code equivalent of the binary number 1010 is
(A) 1101 (B) 1010 (C) 0101 (D) 1111
- Q14.** The expression $A + A'B$ simplifies to
(A) $A + B$ (B) B (C) A (D) AB
- Q15.** A 2-input NOR gate with both inputs tied together behaves as
(A) an OR gate (B) an AND gate (C) an XOR gate (D) a NOT gate
- Q16.** The function $F(A, B) = A \oplus B$ expressed as a sum of minterms is
(A) $\Sigma m(2, 3)$ (B) $\Sigma m(1, 2)$ (C) $\Sigma m(0, 1)$ (D) $\Sigma m(0, 3)$
- Q17.** The number of possible minterms for a Boolean function of 4 variables is
(A) 16 (B) 4 (C) 8 (D) 32
- Q18.** The number of output lines in a 3-to-8 decoder is
(A) 3 (B) 16 (C) 8 (D) 6
- Q19.** The SUM output of a half adder with inputs A and B is
(A) $A \oplus B$ (B) $A \cdot B$ (C) $A + B$ (D) $(A \cdot B)'$
- Q20.** The 2's complement of the 4-bit number 0110 is
(A) 0110 (B) 1001 (C) 1010 (D) 1101
- Q21.** The minimum number of flip-flops required to design a MOD-12 counter is
(A) 4 (B) 12 (C) 3 (D) 6
- Q22.** The race-around condition occurs in a
(A) level-triggered JK flip-flop with $J = K = 1$ (B) SR latch (C) T flip-flop with $T = 0$ (D) D flip-flop
- Q23.** The minimum number of 2-input NAND gates required to implement a 2-input XOR function is
(A) 3 (B) 4 (C) 6 (D) 5
- Q24.** The number of select (control) lines required for a 4-to-1 multiplexer is
(A) 2 (B) 3 (C) 4 (D) 1
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SHEET 3 — COMPUTER ORGANIZATION & ARCHITECTURE

Professional Knowledge — System (CS/IT) | 36 questions | target time 30 minutes

- Q1.** Virtual memory in modern operating systems is normally implemented using
(A) overlays only (B) demand paging (C) contiguous allocation (D) swapping of entire processes only
- Q2.** A memory chip organized as $1K \times 8$ requires
(A) 1024 address lines, 8 data lines (B) 10 address lines, 8 data lines (C) 8 address lines, 10 data lines (D) 10 address lines, 1 data line
- Q3.** The interrupt vector table contains
(A) interrupt data (B) only interrupt priorities (C) starting addresses of interrupt service routines (D) opcodes of interrupt instructions
- Q4.** Translation of a logical address into a physical address at run time is performed by the
(A) ALU (B) compiler (C) DMA controller (D) memory management unit
- Q5.** The register that always holds the address of the next instruction to be fetched is the
(A) Accumulator (B) Program Counter (C) Instruction Register (D) MAR
- Q6.** In a direct-mapped cache with 1024 blocks and a block size of 8 bytes, the number of bits in the block-offset field of the address is
(A) 10 (B) 8 (C) 3 (D) 2
- Q7.** Instruction pipelining primarily improves
(A) the instruction throughput (B) the latency of an individual instruction (C) the size of main memory (D) the clock period only
- Q8.** The stack pointer register always contains
(A) the first instruction of the program (B) the address of the top of the stack (C) the interrupt vector (D) the base of the data segment
- Q9.** The number of address lines required for a 4 KB byte-addressable memory is
(A) 16 (B) 12 (C) 10 (D) 11
- Q10.** The instruction `MOV R1, #5` uses which addressing mode?
(A) Indexed (B) Indirect (C) Direct (D) Immediate
- Q11.** The correct order of operations in a basic instruction cycle is
(A) fetch, decode, execute (B) decode, fetch, execute (C) execute, fetch, decode (D) fetch, execute, decode
- Q12.** The fastest storage element in the memory hierarchy is
(A) CPU registers (B) main memory (C) L1 cache (D) solid-state disk
- Q13.** A 2-way set-associative cache contains 128 blocks in total. The number of sets is
(A) 16 (B) 64 (C) 256 (D) 128
- Q14.** Which I/O technique requires the least CPU involvement per word transferred?
(A) Interrupt-driven I/O (B) Programmed I/O (C) Polling (D) DMA
- Q15.** Zero-address instructions are used in machines organized around
(A) memory-to-memory operations (B) an accumulator (C) general-purpose registers (D) a stack
- Q16.** In a system with a two-level page table and no TLB hit, the number of memory accesses needed to read one data word is
(A) 2 (B) 4 (C) 1 (D) 3
- Q17.** The program counter is normally incremented during the
(A) execute phase of an ALU operation (B) operand decode phase (C) fetch phase (D) interrupt phase only
- Q18.** In a two-address instruction format, the result of an operation is normally placed in
(A) one of the operand locations (B) a dedicated result register only (C) the program counter (D) main memory always
- Q19.** A distinguishing feature of a synchronous bus is
(A) a common clock shared by all devices (B) fully variable transfer timing (C) handshaking signals for every transfer (D) the absence of control lines
- Q20.** A pipeline stall caused by a conditional branch instruction is an example of
(A) control hazard (B) structural hazard (C) data hazard (D) no hazard
- Q21.** A cache has an access time of 10 ns and main memory 100 ns. The hit ratio is 0.95 and on a miss only main memory is accessed. The average memory access time is
(A) 55 ns (B) 14.5 ns (C) 19 ns (D) 10 ns
- Q22.** The cache mapping scheme in which a memory block may be placed in any cache line is
(A) fully associative (B) set associative (C) sector mapped (D) direct mapped
- Q23.** In a little-endian machine, the least significant byte of a multi-byte word is stored at
(A) the lowest address (B) a separate segment (C) the stack top (D) the highest address
- Q24.** A 5-stage pipeline has a stage delay of 2 ns and no hazards. The time to execute 100 instructions is
(A) 210 ns (B) 208 ns (C) 1000 ns (D) 200 ns
- Q25.** A processor supports 64 distinct instructions. The minimum number of bits required for the opcode field is
(A) 7 (B) 8 (C) 6 (D) 5
- Q26.** Memory interleaving is used mainly to improve
(A) cache size (B) memory reliability (C) effective memory bandwidth (D) memory capacity
- Q27.** During a DMA transfer, the system bus is controlled by
(A) the cache controller (B) the interrupt controller (C) the DMA controller (D) the CPU
- Q28.** The addressing mode most suitable for accessing successive elements of an array inside a loop is
(A) Immediate (B) Register (C) Direct (D) Indexed

- Q29.** Under ideal conditions, the speedup obtained from a k-stage instruction pipeline approaches
(A) k^2 (B) $\log k$ (C) $2k$ (D) k
- Q30.** Booth's algorithm is used for
(A) normalization of mantissa (B) multiplication of signed binary numbers (C) floating point addition (D) binary division
- Q31.** A direct-mapped cache of size 32 KB uses a block size of 32 bytes. The number of index bits required is
(A) 15 (B) 10 (C) 5 (D) 8
- Q32.** The throughput of a linear instruction pipeline is determined by
(A) the number of stages alone (B) the slowest stage (C) the fastest stage (D) the average stage delay
- Q33.** The number of address lines required to address 1 MB of byte-addressable memory is
(A) 20 (B) 24 (C) 16 (D) 18
- Q34.** In a write-back cache
(A) main memory is updated only when the block is replaced (B) every write updates main memory immediately (C) a dirty bit is not required (D) writes bypass the cache entirely
- Q35.** Cache memory is normally implemented using
(A) magnetic storage (B) SRAM (C) DRAM (D) EEPROM
- Q36.** Which of the following is a characteristic feature of RISC architecture?
(A) Complex microprogrammed control for most instructions (B) A very large number of addressing modes (C) Fixed-length instructions (D) Variable-length instructions
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SHEET 4 — PROGRAMMING & DATA STRUCTURES

Professional Knowledge — System (CS/IT) | 36 questions | target time 30 minutes

- Q1.** Which pair of traversals does NOT uniquely determine a binary tree?
(A) Preorder and postorder (B) Inorder and level order (C) Inorder and preorder (D) Inorder and postorder
- Q2.** In the array representation of a binary heap with the root at index 1, the children of the node at index i are at indices
(A) $2i - 1$ and $2i$ (B) $2i$ and $2i + 1$ (C) $i/2$ and $i/2 + 1$ (D) $i - 1$ and $i + 1$
- Q3.** The most space-efficient way to store a sparse matrix is
(A) a circular queue (B) a two-dimensional array (C) a triplet or linked representation of non-zero elements (D) a stack
- Q4.** In C, when an array is passed as an argument to a function, what is actually passed is
(A) a pointer to its first element (B) a copy of the entire array (C) a reference to the whole array object (D) the array size only
- Q5.** A circular queue implemented in an array of size n , keeping one slot always empty to distinguish full from empty, can hold at most
(A) $n - 1$ elements (B) $n/2$ elements (C) $n + 1$ elements (D) n elements
- Q6.** Inserting the sorted sequence 1, 2, 3, ..., n into an empty binary search tree produces a tree of height
(A) $\sqrt{n}$ (B) $\log_2 n$ (C) $n - 1$ (D) $n/2$
- Q7.** The minimum number of stacks required to implement a queue is
(A) 2 (B) 4 (C) 1 (D) 3
- Q8.** Depth-first search of a graph is naturally implemented using
(A) a stack (B) a priority queue (C) a hash table (D) a queue
- Q9.** In C, for the declaration `int a[5];` the expression `a[2]` is equivalent to
(A) `*(a + 2)` (B) `*a + 2` (C) `a + 2` (D) `&a + 2`
- Q10.** In C, the value of `sizeof(char)` is guaranteed to be
(A) machine dependent (B) 2 (C) 1 (D) 4
- Q11.** Given a pointer to a node in a doubly linked list, the time to delete that node is
(A) $O(n^2)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(1)$
- Q12.** To achieve $O(1)$ enqueue and $O(1)$ dequeue in a linked-list-based queue, the implementation must maintain
(A) only a head pointer (B) only a tail pointer (C) a doubly linked list, necessarily (D) both head and tail pointers
- Q13.** A queue implemented in a linear array with simple front and rear indices suffers from
(A) loss of FIFO order (B) false overflow even when slots are free (C) inability to store integers (D) $O(n)$ enqueue time
- Q14.** The most efficient data structure for implementing a priority queue supporting $O(\log n)$ insert and extract-min is
(A) a stack (B) a sorted array (C) a binary heap (D) an unsorted linked list
- Q15.** Which data structure is used for the evaluation of a postfix expression?
(A) Queue (B) Priority queue (C) Circular queue (D) Stack
- Q16.** In a threaded binary tree, NULL pointers are used to
(A) point to the in-order predecessor or successor (B) reduce the height of the tree (C) balance the tree (D) store additional data
- Q17.** Iterative reversal of a singly linked list requires a minimum of how many pointer variables?
(A) 1 (B) 4 (C) 2 (D) 3
- Q18.** Linear probing as a collision resolution technique suffers from
(A) chaining overhead (B) no clustering at all (C) secondary clustering (D) primary clustering
- Q19.** The minimum number of nodes in an AVL tree of height 3 (root at height 0) is
(A) 6 (B) 8 (C) 15 (D) 7
- Q20.** The time to search for a key in a height-balanced binary search tree with n nodes is
(A) $O(n)$ (B) $O(\log n)$ (C) $O(n \log n)$ (D) $O(1)$
- Q21.** The number of distinct binary search trees that can be formed with 3 distinct keys is
(A) 5 (B) 6 (C) 3 (D) 9
- Q22.** The postorder traversal of an expression tree produces the expression in
(A) prefix form (B) infix form (C) postfix form (D) level order
- Q23.** The height of a complete binary tree with n nodes (root at height 0) is
(A) $n/2$ (B) $\lceil n/2 \rceil$ (C) $\lceil \log_2 n \rceil$ (D) $\log_2(n + 1)$
- Q24.** The worst-case time to insert a node at the beginning of a singly linked list with n nodes is
(A) $O(n)$ (B) $O(n \log n)$ (C) $O(1)$ (D) $O(\log n)$
- Q25.** The worst-case height of a binary search tree containing n nodes is
(A) $\log_2 n$ (B) $\sqrt{n}$ (C) $n - 1$ (D) $n/2$
- Q26.** Elements 1, 2, 3 are pushed onto a stack in that order, with pops interleaved arbitrarily. Which output sequence is NOT possible?
(A) 3, 2, 1 (B) 3, 1, 2 (C) 1, 3, 2 (D) 2, 3, 1
- Q27.** In a full binary tree in which every internal node has exactly 2 children, if there are 10 leaf nodes then the total number of nodes is
(A) 20 (B) 18 (C) 21 (D) 19
- Q28.** In a C program, memory obtained through `malloc()` is allocated from the
(A) heap (B) code segment (C) data segment (D) stack

- Q29.** The in-order traversal of a binary search tree yields the keys in
(A) sorted ascending order (B) random order (C) level order (D) reverse sorted order
- Q30.** On a machine where pointers are 64 bits wide, the value of `sizeof(int *)` in C is
(A) 8 (B) equal to `sizeof(int)` (C) 2 (D) 4
- Q31.** A recursive function written without a base case will most likely cause
(A) a compile-time error (B) heap fragmentation only (C) a run-time stack overflow (D) an infinite loop using no extra memory
- Q32.** The level-order traversal of a binary tree is implemented using
(A) a BST (B) a max-heap (C) a stack (D) a queue
- Q33.** A binary tree with n nodes has how many NULL child pointers?
(A) $n + 1$ (B) $n - 1$ (C) n (D) $2n$
- Q34.** The maximum number of nodes in a binary tree of height h (root at height 0) is
(A) $2h + 1$ (B) 2^h (C) $2^h - 1$ (D) $2^{(h+1)} - 1$
- Q35.** In C, the value returned by `strlen("hello")` is
(A) 4 (B) 5 (C) 6 (D) undefined
- Q36.** The postfix form of the expression $(A + B) * (C - D)$ is
(A) $+AB-CD*$ (B) $AB+CD-*$ (C) $ABCD+-*$ (D) $AB+C-D*$
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SHEET 5 — ALGORITHMS

Professional Knowledge — System (CS/IT) | 36 questions | target time 30 minutes

- Q1.** Which of the following employ the divide-and-conquer strategy?
(A) Binary search only (B) Merge sort only (C) Quicksort only (D) All of merge sort, binary search and quicksort
- Q2.** The time complexity of the Bellman-Ford algorithm on a graph with V vertices and E edges is
(A) $O(VE)$ (B) $O(V^2)$ (C) $O(E \log V)$ (D) $O(V + E)$
- Q3.** Strassen's algorithm multiplies two $n \times n$ matrices in approximately
(A) $O(n^{2.81})$ (B) $O(n^3)$ (C) $O(n^2)$ (D) $O(n \log n)$
- Q4.** The minimum number of comparisons required to find both the minimum and the maximum of n elements is
(A) $2n - 2$ (B) $n - 1$ (C) $n \log n$ (D) $\lceil 3n/2 \rceil - 2$
- Q5.** Which algorithm selects the k th smallest element of an unsorted array in $O(n)$ worst-case time?
(A) Merge sort (B) Heapsort (C) Quickselect with a random pivot (D) Median-of-medians selection
- Q6.** The average-case time complexity of quicksort is
(A) $O(n \log n)$ (B) $O(n^2)$ (C) $O(n)$ (D) $O(\log n)$
- Q7.** The auxiliary space complexity of the standard merge sort on arrays is
(A) $O(\log n)$ (B) $O(n \log n)$ (C) $O(n)$ (D) $O(1)$
- Q8.** Matrix chain multiplication is solved optimally using
(A) divide and conquer (B) greedy method (C) dynamic programming (D) branch and bound
- Q9.** Which sorting algorithm runs in $O(n)$ time when the input consists of integers drawn from a small fixed range?
(A) Counting sort (B) Quicksort (C) Merge sort (D) Heapsort
- Q10.** The greedy approach does NOT yield an optimal solution for
(A) the fractional knapsack problem (B) the 0/1 knapsack problem (C) minimum spanning tree (D) Huffman coding
- Q11.** A topological ordering of vertices exists for
(A) directed graphs with cycles (B) any graph (C) any undirected graph (D) directed acyclic graphs only
- Q12.** The dynamic programming solution for the longest common subsequence of strings of lengths m and n takes
(A) $O(m + n)$ time (B) $O(n \log n)$ time (C) $O(mn)$ time (D) $O(2^n)$ time
- Q13.** Dijkstra's single-source shortest path algorithm may produce incorrect results when the graph has
(A) more than one source (B) negative edge weights (C) parallel edges (D) cycles
- Q14.** Which algorithm computes single-source shortest paths correctly in a graph having negative edge weights but no negative cycle?
(A) Bellman-Ford algorithm (B) Kruskal's algorithm (C) Dijkstra's algorithm (D) Prim's algorithm
- Q15.** The fractional knapsack problem is optimally solved by
(A) branch and bound (B) backtracking (C) the greedy method (D) dynamic programming
- Q16.** The time complexity of building a binary heap from an unsorted array of n elements using the bottom-up procedure is
(A) $O(n^2)$ (B) $O(n \log n)$ (C) $O(\log n)$ (D) $O(n)$
- Q17.** In a hash table with n keys, m slots and collision resolution by separate chaining, the average time for an unsuccessful search is
(A) $O(1 + n/m)$ (B) $O(1)$ (C) $O(n)$ (D) $O(\log n)$
- Q18.** The Floyd-Warshall all-pairs shortest path algorithm has time complexity
(A) $O(V^2)$ (B) $O(E \log V)$ (C) $O(VE)$ (D) $O(V^3)$
- Q19.** Binary search cannot be applied efficiently to a sorted singly linked list because
(A) comparisons are costlier (B) recursion is not allowed (C) accessing the middle element takes $O(n)$ time (D) the list is unsorted
- Q20.** Which of the following functions grows the fastest asymptotically?
(A) $n \log n$ (B) n^2 (C) 2^n (D) $n!$
- Q21.** Kruskal's algorithm for minimum spanning tree is based on which design technique?
(A) Backtracking (B) Dynamic programming (C) Greedy (D) Divide and conquer
- Q22.** The greedy activity-selection algorithm obtains an optimal solution by repeatedly choosing the activity with the
(A) highest value (B) shortest duration (C) earliest finishing time (D) earliest starting time
- Q23.** The worst-case time complexity of heapsort is
(A) $O(n \log n)$ (B) $O(n^2)$ (C) $O(\log n)$ (D) $O(n)$
- Q24.** Huffman coding is
(A) a greedy but sub-optimal algorithm (B) a greedy algorithm that produces an optimal prefix code (C) a dynamic programming algorithm producing fixed-length codes (D) a divide-and-conquer algorithm
- Q25.** Quicksort is applied to an already sorted array of n elements with the first element chosen as pivot. The running time is
(A) $O(n)$ (B) $O(\log n)$ (C) $O(n \log n)$ (D) $O(n^2)$
- Q26.** The maximum number of swaps performed by selection sort on n elements is
(A) $n - 1$ (B) n^2 (C) $n^2/2$ (D) $n \log n$
- Q27.** Which of the following sorting algorithms is stable?
(A) Heapsort (B) Selection sort (C) Quicksort (D) Merge sort
- Q28.** Kruskal's algorithm, implemented with sorting and union-find, runs in time
(A) $O(V^2)$ (B) $O(V^3)$ (C) $O(E \log E)$ (D) $O(E + V)$

- Q29.** The solution of $T(n) = 2T(n/2) + n$, $T(1) = 1$ is
(A) $\Theta(n)$ (B) $\Theta(\log n)$ (C) $\Theta(n \log n)$ (D) $\Theta(n^2)$
- Q30.** Merging two sorted arrays of sizes m and n requires, in the worst case,
(A) $m + n - 1$ comparisons (B) mn comparisons (C) $\max(m, n)$ comparisons (D) $m + n$ comparisons
- Q31.** The solution of the recurrence $T(n) = 4T(n/2) + n^2$ is
(A) $\Theta(n^2)$ (B) $\Theta(n \log n)$ (C) $\Theta(n^2 \log n)$ (D) $\Theta(n^3)$
- Q32.** The maximum number of comparisons made by binary search on a sorted array of 1000 elements is
(A) 999 (B) 100 (C) 500 (D) 10
- Q33.** Prim's algorithm implemented with an adjacency matrix and no priority queue runs in
(A) $O(V + E)$ (B) $O(V^2)$ (C) $O(V^3)$ (D) $O(E \log V)$
- Q34.** The best-case time complexity of insertion sort is
(A) $O(n^2)$ (B) $O(n)$ (C) $O(n \log n)$ (D) $O(1)$
- Q35.** The solution of the recurrence $T(n) = T(n/2) + 1$ is
(A) $\Theta(\log n)$ (B) $\Theta(n \log n)$ (C) $\Theta(n)$ (D) $\Theta(1)$
- Q36.** The worst-case number of comparisons in a linear search over n elements is
(A) n (B) $n/2$ (C) $n \log n$ (D) $\log n$
-

SHEET 6 — THEORY OF COMPUTATION

Professional Knowledge — System (CS/IT) | 24 questions | target time 20 minutes

- Q1.** The regular expression denoting all strings of even length over the alphabet $\{a\}$ is
(A) $a(aa)^*$ (B) $(a + aa)^*$ (C) $(aa)^*$ (D) a^*
- Q2.** The pumping lemma for regular languages is used to prove that a given language is
(A) decidable (B) not regular (C) context-free (D) regular
- Q3.** The halting problem for Turing machines is
(A) undecidable but recursively enumerable (B) decidable (C) not recursively enumerable (D) context-free
- Q4.** Context-free languages are closed under
(A) complementation (B) union (C) intersection (D) all of the above
- Q5.** The problem of deciding whether the language generated by a given context-free grammar is empty is
(A) not recursively enumerable (B) decidable (C) NP-complete but undecidable (D) undecidable
- Q6.** A context-free grammar is said to be ambiguous if
(A) it has useless productions (B) it generates an infinite language (C) some string has more than one parse tree (D) it is left recursive
- Q7.** The number of states in the minimal DFA for the regular expression $(0 + 1)^*11$ is
(A) 5 (B) 2 (C) 3 (D) 4
- Q8.** The intersection of a context-free language with a regular language is always
(A) not context-free (B) context-free (C) finite (D) regular
- Q9.** Which of the following containments is correct?
(A) $\text{Regular} \subset \text{CFL} \subset \text{Recursive}$ (B) $\text{Regular} = \text{CFL}$ (C) $\text{Recursive} \subset \text{CFL}$ (D) $\text{CFL} \subset \text{Regular}$
- Q10.** Regarding the classes of languages accepted by DFA and NFA
(A) the two classes are incomparable (B) both accept exactly the same class (C) DFAs accept strictly more languages (D) NFAs accept strictly more languages
- Q11.** The minimum number of states in a DFA over $\{0, 1\}$ accepting all strings ending with 01 is
(A) 4 (B) 2 (C) 3 (D) 5
- Q12.** The language of all palindromes over $\{a, b\}$ is
(A) not context-free (B) regular (C) context-free but not regular (D) finite
- Q13.** The tape of a standard Turing machine is
(A) read-only (B) unbounded in length (C) finite and fixed (D) write-only
- Q14.** The minimum number of states in a DFA accepting the language of the regular expression $(a + b)^*abb$ is
(A) 5 (B) 4 (C) 6 (D) 3
- Q15.** The language $L = \{w \in \{a, b\}^* \mid w \text{ has an equal number of } a\text{'s and } b\text{'s}\}$ is
(A) not context-free (B) finite (C) context-free but not regular (D) regular
- Q16.** The minimum number of states in a DFA over $\{a, b\}$ that accepts all strings in which the number of a 's is divisible by 3 is
(A) 4 (B) 2 (C) 6 (D) 3
- Q17.** Regular languages are closed under
(A) intersection only (B) complementation only (C) union, intersection and complementation (D) union only
- Q18.** In the Chomsky hierarchy, a Type-3 grammar generates
(A) regular languages (B) context-free languages (C) context-sensitive languages (D) recursively enumerable languages
- Q19.** Deterministic pushdown automata accept
(A) all context-sensitive languages (B) all context-free languages (C) a proper subset of the context-free languages (D) only regular languages
- Q20.** The language $L = \{a^n b^n \mid n \geq 1\}$ is
(A) finite (B) context-free but not regular (C) regular (D) not context-free
- Q21.** A language accepted by a Turing machine that halts on every input is
(A) recursive (decidable) (B) necessarily context-free (C) recursively enumerable but not recursive (D) not recursively enumerable
- Q22.** Which class of languages is NOT closed under complementation?
(A) Finite languages (B) Recursive languages (C) Context-free languages (D) Regular languages
- Q23.** The grammar $S \rightarrow aSb \mid \epsilon$ generates the language
(A) $\{a^n b^m \mid n, m \geq 0\}$ (B) $(ab)^*$ (C) $\{a^n b^n \mid n \geq 0\}$ (D) all strings over $\{a, b\}$
- Q24.** In the worst case, the number of states in a DFA equivalent to an n -state NFA can be as large as
(A) n^2 (B) $n!$ (C) n (D) 2^n
-

SHEET 7 — COMPILER DESIGN

Professional Knowledge — System (CS/IT) | 24 questions | target time 20 minutes

- Q1.** Peephole optimization is performed on
(A) the parse tree (B) a small sliding window of target code (C) the source program (D) the symbol table
- Q2.** Which parser is the most powerful among the following?
(A) CLR(1) (B) LALR(1) (C) SLR(1) (D) LL(1)
- Q3.** Operator precedence parsing is a form of
(A) predictive parsing (B) recursive descent parsing (C) shift-reduce (bottom-up) parsing (D) top-down parsing
- Q4.** The DAG representation of a basic block is useful for
(A) tokenizing (B) common sub-expression elimination (C) register spilling only (D) parsing
- Q5.** The symbol table in a compiler is
(A) used by almost all phases of the compiler (B) created only during code generation (C) discarded after parsing (D) built only during lexical analysis
- Q6.** Assigning a real value to an integer variable without a cast is detected during
(A) semantic analysis (B) syntax analysis (C) code generation (D) lexical analysis
- Q7.** YACC is a tool that generates a
(A) assembler (B) lexical analyzer (C) parser (D) code optimizer
- Q8.** An S-attributed syntax-directed definition uses
(A) no attributes (B) both, evaluated left to right (C) only synthesized attributes (D) only inherited attributes
- Q9.** FIRST and FOLLOW sets are computed for the purpose of
(A) register allocation (B) constructing an LL(1) parsing table (C) peephole optimization (D) lexical analysis
- Q10.** Register allocation and assignment is performed during
(A) code generation (B) lexical analysis (C) semantic analysis (D) parsing
- Q11.** Moving a computation that does not change across iterations out of a loop is called
(A) constant folding (B) loop-invariant code motion (C) strength reduction (D) dead code elimination
- Q12.** The output of the lexical analysis phase is
(A) a stream of tokens (B) intermediate code (C) a parse tree (D) a symbol table only
- Q13.** A basic block in intermediate code is a sequence of instructions with
(A) a single entry point and a single exit point (B) at least one loop (C) multiple entry points (D) exactly one instruction
- Q14.** Which of the following is a machine-independent optimization?
(A) Instruction selection (B) Peephole optimization (C) Common sub-expression elimination (D) Register allocation
- Q15.** Handle pruning is a technique associated with
(A) top-down parsing (B) semantic analysis (C) bottom-up parsing (D) lexical analysis
- Q16.** The minimum number of temporary variables needed in the three-address code for $a = b * c + d * e$ is
(A) 1 (B) 4 (C) 3 (D) 2
- Q17.** Left factoring of a grammar is performed to make it suitable for
(A) LR parsing (B) predictive (LL(1)) parsing (C) lexical analysis (D) peephole optimization
- Q18.** The compiler phase that constructs the parse tree is
(A) semantic analysis (B) code optimization (C) syntax analysis (D) lexical analysis
- Q19.** A recursive descent parser is an example of a
(A) bottom-up parser (B) top-down parser (C) LR parser (D) operator precedence parser
- Q20.** Elimination of left recursion from a grammar is essential for
(A) code optimization (B) LR parsing (C) lexical analysis (D) top-down parsing
- Q21.** Constant folding is an optimization performed
(A) by the operating system (B) at run time only (C) during linking (D) at compile time
- Q22.** An activation record created on the run-time stack typically contains
(A) only global variables (B) local variables, parameters and the return address (C) generated machine code (D) the complete symbol table
- Q23.** An L-attributed syntax-directed definition permits inherited attributes to be computed from
(A) the parent and left siblings (B) no other nodes (C) right siblings only (D) children only
- Q24.** An LR parser uses which data structure to hold states and symbols?
(A) Heap (B) Queue (C) Stack (D) Hash table
-

SHEET 8 — OPERATING SYSTEM

Professional Knowledge — System (CS/IT) | 36 questions | target time 30 minutes

- Q1.** Which CPU scheduling algorithm is preemptive by design and gives every process a fixed time slice?
(A) Round Robin (B) SJF (C) FCFS (D) Priority (non-preemptive)
- Q2.** A race condition occurs when
(A) a process waits indefinitely for a resource (B) processes never share data (C) the final result depends on the relative order of accesses to shared data (D) CPU utilization reaches 100%
- Q3.** Thrashing in a virtual memory system is directly caused by
(A) large page size (B) contiguous allocation (C) an excessively high page fault rate (D) too few processes in memory
- Q4.** Which disk scheduling algorithm can cause starvation of requests located far from the current head position?
(A) SCAN (B) SSTF (C) C-SCAN (D) FCFS
- Q5.** In a system with a 32-bit logical address space and 4 KB pages, the number of pages in the logical address space is
(A) 2^{20} (B) 2^{10} (C) 2^{32} (D) 2^{12}
- Q6.** Which scheduling policy can cause indefinite postponement (starvation) of long jobs?
(A) First-in first-out with aging (B) Round Robin (C) FCFS (D) Shortest Job First
- Q7.** A counting semaphore initialized to 1 and used with wait/signal around a critical section behaves as
(A) a spin lock only (B) a condition variable (C) a monitor (D) a binary semaphore / mutex
- Q8.** A zombie process is one that
(A) has no parent process (B) is permanently blocked on a semaphore (C) is waiting for an I/O device (D) has terminated but whose exit status has not yet been collected by its parent
- Q9.** The problem of priority inversion in real-time systems is addressed by
(A) aging (B) priority inheritance (C) FCFS scheduling (D) round robin scheduling
- Q10.** In which threading model does the blocking of one thread NOT block the entire process?
(A) Pure user-level threads (B) Kernel-level threads (C) Neither model (D) Both behave identically
- Q11.** Belady's anomaly is exhibited by which page replacement policy?
(A) LFU (B) FIFO (C) LRU (D) Optimal
- Q12.** Four processes arrive at time 0 with CPU bursts 6, 8, 7 and 3 units. Under SJF scheduling the average waiting time is
(A) 10.25 units (B) 7 units (C) 6.5 units (D) 8 units
- Q13.** With a page size of 1 KB and a 16-bit logical address, the number of pages in the logical address space is
(A) 16 (B) 1024 (C) 64 (D) 32
- Q14.** Which of the following is NOT one of the four necessary conditions for deadlock?
(A) Mutual exclusion (B) Circular wait (C) Hold and wait (D) Priority inversion
- Q15.** A process sitting in the ready queue is waiting for
(A) the CPU (B) a semaphore signal (C) a page to be loaded (D) an I/O device
- Q16.** External fragmentation is a problem primarily in
(A) variable-partition contiguous allocation (B) demand paging (C) TLB management (D) paging
- Q17.** The Translation Lookaside Buffer (TLB) is
(A) part of secondary storage (B) a page replacement policy (C) a cache of recently used page table entries (D) a file allocation index
- Q18.** Spinlocks are the preferred synchronization mechanism when
(A) critical sections are very long (B) critical sections are very short and multiple processors are available (C) there is a single CPU and long critical sections (D) processes never contend for the lock
- Q19.** The convoy effect is associated with which scheduling algorithm?
(A) FCFS (B) SJF (C) Multilevel feedback queue (D) Round Robin
- Q20.** Which of the following is NOT a requirement of a correct critical-section solution?
(A) Bounded waiting (B) Mutual exclusion (C) Progress (D) Assumptions about relative process speeds
- Q21.** In UNIX/Linux, the system call used to create a new process is
(A) exec() (B) wait() (C) fork() (D) pipe()
- Q22.** Which CPU scheduling algorithm gives the minimum average waiting time for a given set of processes arriving together?
(A) Priority (B) Shortest Job First (C) Round Robin (D) FCFS
- Q23.** For the reference string 1, 2, 3, 4, 1, 2, 5 with 3 frames and FIFO replacement, the number of page faults is
(A) 6 (B) 7 (C) 5 (D) 4
- Q24.** The optimal (OPT/MIN) page replacement algorithm
(A) gives the lowest possible fault rate but is not implementable in practice (B) suffers from Belady's anomaly (C) behaves identically to FIFO (D) is easily implementable
- Q25.** As the time quantum in Round Robin scheduling is made very small,
(A) context switch overhead increases (B) it behaves like SJF (C) starvation of long processes increases (D) context switch overhead decreases
- Q26.** Internal fragmentation is an inherent problem of
(A) pure segmentation (B) paging (C) both equally (D) neither
- Q27.** A process executes the statements `fork(); fork();`. The total number of processes in existence (including the original) is
(A) 3 (B) 4 (C) 8 (D) 2

- Q28.** The working set model is used primarily to
(A) speed up CPU scheduling (B) improve cache hit ratio (C) allocate disk blocks (D) prevent thrashing
- Q29.** The standard semaphore solution to the bounded-buffer producer-consumer problem uses how many semaphores?
(A) 1 (B) 2 (C) 4 (D) 3
- Q30.** The Banker's algorithm is used for
(A) deadlock detection (B) deadlock avoidance (C) page replacement (D) deadlock recovery
- Q31.** A transition of a process from the running state to the ready state is caused by
(A) process termination (B) an I/O request (C) a page fault (D) expiry of the time quantum
- Q32.** Which of the following is NOT an inter-process communication mechanism?
(A) Paging (B) Pipes (C) Shared memory (D) Message passing
- Q33.** Contiguous allocation of disk files provides
(A) no fragmentation at all (B) mandatory use of an index block (C) slow direct access (D) fast direct access but suffers from external fragmentation
- Q34.** In a system where every resource has exactly one instance, deadlock is detected by
(A) finding a cycle in the wait-for graph (B) checking the ready queue length (C) running the Banker's algorithm (D) counting page faults
- Q35.** The operating system module that performs the actual switching of the CPU to the selected process is the
(A) loader (B) long-term scheduler (C) linker (D) dispatcher
- Q36.** Mutual exclusion implemented using the TestAndSet instruction is an example of
(A) a compiler directive (B) a purely software solution (C) a hardware-supported solution involving busy waiting (D) interrupt disabling only
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SHEET 9 — DATABASES

Professional Knowledge — System (CS/IT) | 24 questions | target time 20 minutes

- Q1.** Which ACID property guarantees that the effects of a committed transaction survive subsequent system failures?
(A) Consistency (B) Atomicity (C) Durability (D) Isolation
- Q2.** Deadlock prevention in database concurrency control is achieved by schemes such as
(A) shadow paging (B) checkpointing (C) two-phase commit (D) wait-die and wound-wait
- Q3.** A trigger in a relational database is
(A) an index structure (B) a procedure executed automatically in response to a specified event (C) a constraint checked manually (D) a virtual table
- Q4.** Which of the following is a DDL command?
(A) DELETE (B) UPDATE (C) CREATE (D) SELECT
- Q5.** A primary index built on the ordering key field of an ordered data file is normally
(A) a secondary index (B) always multilevel (C) sparse (D) dense
- Q6.** $R(A, B, C, D)$ has the FD set $\{AB \rightarrow C, C \rightarrow D, D \rightarrow A\}$. The highest normal form satisfied by R is
(A) 2NF (B) BCNF (C) 3NF (D) 1NF
- Q7.** For $R(A, B, C)$ with the FD set $\{A \rightarrow B, B \rightarrow C\}$, the candidate key is
(A) A (B) C (C) BC (D) B
- Q8.** A relation is in first normal form (1NF) if
(A) every attribute value is atomic (B) it has no transitive dependency (C) every determinant is a candidate key (D) it has no partial dependency
- Q9.** Two-phase locking (2PL) protocol guarantees
(A) deadlock freedom (B) recoverability only (C) no starvation (D) conflict serializability
- Q10.** In a B+ tree index, pointers to actual data records are stored in
(A) leaf nodes only (B) the root only (C) internal nodes only (D) both internal and leaf nodes
- Q11.** In SQL, $\text{COUNT}(\text{column_name})$
(A) is always equal to $\text{COUNT}(\ast)$ (B) counts NULL values as well (C) is invalid syntax (D) ignores NULL values
- Q12.** A schedule is conflict serializable if and only if its precedence (serializability) graph
(A) is acyclic (B) is bipartite (C) is a complete graph (D) contains a cycle
- Q13.** If relation R has 5 tuples and relation S has 4 tuples, the number of tuples in $R \times S$ is
(A) 4 (B) 9 (C) 5 (D) 20
- Q14.** The projection operation in relational algebra
(A) retains duplicate tuples (B) selects rows only (C) always returns one tuple (D) eliminates duplicate tuples
- Q15.** In SQL, the clause used to filter groups produced by GROUP BY is
(A) ORDER BY (B) HAVING (C) WHERE (D) DISTINCT
- Q16.** Which SQL set operator eliminates duplicate rows by default?
(A) CROSS JOIN (B) UNION ALL (C) JOIN (D) UNION
- Q17.** Which statement about TRUNCATE is correct?
(A) It is a DDL command that removes all rows of a table (B) It drops the table structure as well (C) It is a DML command that can be rolled back easily (D) It deletes selected rows using a WHERE clause
- Q18.** A view in SQL is
(A) a backup file (B) a virtual table derived from a query (C) an index (D) a physically stored table
- Q19.** Referential integrity in a relational database is enforced through
(A) a check constraint (B) a unique constraint (C) a foreign key (D) a primary key
- Q20.** $R(A, B, C, D)$ has A as its only candidate key. The number of superkeys of R is
(A) 4 (B) 16 (C) 8 (D) 15
- Q21.** Which of the following is NOT a primitive (basic) operation of relational algebra?
(A) Projection (B) Union (C) Selection (D) Natural join
- Q22.** A decomposition of R into R_1 and R_2 is lossless if
(A) both are in BCNF (B) $R_1 \cap R_2$ is empty (C) the common attributes form a superkey of at least one of R_1 or R_2 (D) $R_1 \cup R_2 \neq R$
- Q23.** A schedule is said to be serializable if it is equivalent to
(A) a schedule with no read operations (B) any schedule without aborts (C) a concurrent schedule with locks (D) some serial schedule of the same transactions
- Q24.** The minimum possible number of tuples in the natural join of two non-empty relations is
(A) 0 (B) the product of the two sizes (C) the size of the smaller relation (D) 1
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SHEET 10 — COMPUTER NETWORKS

Professional Knowledge — System (CS/IT) | 24 questions | target time 20 minutes

- Q1.** Cyclic Redundancy Check (CRC) is used for
(A) encryption (B) routing (C) flow control (D) error detection
- Q2.** The number of usable host addresses in the subnet 192.168.10.0/26 is
(A) 62 (B) 126 (C) 30 (D) 64
- Q3.** IEEE 802.11 wireless LANs use which medium access method?
(A) Pure ALOHA (B) CSMA/CA (C) CSMA/CD (D) Token ring
- Q4.** The length of an IPv6 address is
(A) 64 bits (B) 128 bits (C) 256 bits (D) 32 bits
- Q5.** Which of the following normally uses UDP at the transport layer?
(A) HTTP (B) DNS query (C) SMTP (D) FTP control connection
- Q6.** Which transport layer protocol is connectionless?
(A) FTP (B) TCP (C) SCTP (D) UDP
- Q7.** The ping utility works using which protocol?
(A) IGMP (B) ARP (C) SNMP (D) ICMP
- Q8.** Fragmentation of IP datagrams is performed at which layer?
(A) Application layer (B) Transport layer (C) Network layer (D) Data link layer
- Q9.** Which DNS resource record maps a domain name to an IPv4 address?
(A) PTR (B) CNAME (C) A (D) MX
- Q10.** The number of /27 subnets that can be carved out of a single /24 network is
(A) 4 (B) 32 (C) 8 (D) 16
- Q11.** The number of usable host addresses in a /30 subnet is
(A) 2 (B) 4 (C) 1 (D) 6
- Q12.** The layer responsible for reliable end-to-end (process-to-process) delivery is the
(A) network layer (B) transport layer (C) data link layer (D) session layer
- Q13.** The correct sequence of segments in the TCP connection establishment handshake is
(A) ACK, SYN, SYN+ACK (B) SYN, SYN+ACK, ACK (C) SYN, FIN, ACK (D) SYN, ACK, SYN
- Q14.** A LAN switch operates primarily at which OSI layer?
(A) Network (B) Data link (C) Physical (D) Transport
- Q15.** A sliding window protocol with a sender window size of 1 reduces to
(A) Stop-and-Wait (B) Go-Back-N (C) Selective Repeat (D) piggybacked sliding window
- Q16.** The default subnet mask of a Class B IPv4 network is
(A) 255.0.0.0 (B) 255.255.0.0 (C) 255.255.255.0 (D) 255.255.255.255
- Q17.** The well-known default port number for HTTP is
(A) 80 (B) 21 (C) 443 (D) 25
- Q18.** Distance vector routing suffers from which problem?
(A) Hidden terminal (B) Count-to-infinity (C) Silly window syndrome (D) Flooding storm
- Q19.** The TTL field in an IPv4 header prevents
(A) congestion (B) packets from circulating indefinitely (C) fragmentation (D) duplicate acknowledgements
- Q20.** According to the Nyquist criterion, the maximum data rate of a noiseless channel of bandwidth H with V discrete signal levels is
(A) $H \log_2 V$ (B) $2H \log_2 V$ (C) $H \log_2 (1 + S/N)$ (D) $2H V$
- Q21.** The well-known port number used by SMTP is
(A) 25 (B) 23 (C) 21 (D) 110
- Q22.** In Go-Back-N ARQ using 3-bit sequence numbers, the maximum sender window size is
(A) 3 (B) 4 (C) 8 (D) 7
- Q23.** Network Address Translation (NAT) operates at which layer?
(A) Data link (B) Application (C) Network (D) Session
- Q24.** The Address Resolution Protocol (ARP) maps
(A) an IP address to a MAC address (B) a domain name to an IP address (C) a port number to a process (D) a MAC address to an IP address
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ANSWER KEYS

Sheet 1 — Engineering Mathematics: 1-C 2-C 3-C 4-B 5-A 6-A 7-A 8-C 9-D 10-A 11-D 12-C 13-C 14-D 15-D 16-B 17-C 18-A 19-A 20-D 21-D 22-B 23-D 24-C 25-D 26-A 27-A 28-B 29-D 30-D 31-A 32-A 33-D 34-A 35-D 36-D

Sheet 2 — Digital Logic: 1-B 2-D 3-A 4-D 5-A 6-C 7-B 8-A 9-A 10-D 11-C 12-D 13-D 14-A 15-D 16-B 17-A 18-C 19-A 20-C 21-A 22-A 23-B 24-A

Sheet 3 — Computer Organization & Architecture: 1-B 2-B 3-C 4-D 5-B 6-C 7-A 8-B 9-B 10-D 11-A 12-A 13-B 14-D 15-D 16-D 17-C 18-A 19-A 20-A 21-B 22-A 23-A 24-B 25-C 26-C 27-C 28-D 29-D 30-B 31-B 32-B 33-A 34-A 35-B 36-C

Sheet 4 — Programming & Data Structures: 1-A 2-B 3-C 4-A 5-A 6-C 7-A 8-A 9-A 10-C 11-D 12-D 13-B 14-C 15-D 16-A 17-D 18-D 19-D 20-B 21-A 22-C 23-C 24-C 25-C 26-B 27-D 28-A 29-A 30-A 31-C 32-D 33-A 34-D 35-B 36-B

Sheet 5 — Algorithms: 1-D 2-A 3-A 4-D 5-D 6-A 7-C 8-C 9-A 10-B 11-D 12-C 13-B 14-A 15-C 16-D 17-A 18-D 19-C 20-D 21-C 22-C 23-A 24-B 25-D 26-A 27-D 28-C 29-C 30-A 31-C 32-D 33-B 34-B 35-A 36-A

Sheet 6 — Theory of Computation: 1-C 2-B 3-A 4-B 5-B 6-C 7-C 8-B 9-A 10-B 11-C 12-C 13-B 14-B 15-C 16-D 17-C 18-A 19-C 20-B 21-A 22-C 23-C 24-D

Sheet 7 — Compiler Design: 1-B 2-A 3-C 4-B 5-A 6-A 7-C 8-C 9-B 10-A 11-B 12-A 13-A 14-C 15-C 16-C 17-B 18-C 19-B 20-D 21-D 22-B 23-A 24-C

Sheet 8 — Operating System: 1-A 2-C 3-C 4-B 5-A 6-D 7-D 8-D 9-B 10-B 11-B 12-B 13-C 14-D 15-A 16-A 17-C 18-B 19-A 20-D 21-C 22-B 23-B 24-A 25-A 26-B 27-B 28-D 29-D 30-B 31-D 32-A 33-D 34-A 35-D 36-C

Sheet 9 — Databases: 1-C 2-D 3-B 4-C 5-C 6-C 7-A 8-A 9-D 10-A 11-D 12-A 13-D 14-D 15-B 16-D 17-A 18-B 19-C 20-C 21-D 22-C 23-D 24-A

Sheet 10 — Computer Networks: 1-D 2-A 3-B 4-B 5-B 6-D 7-D 8-C 9-C 10-C 11-A 12-B 13-B 14-B 15-A 16-B 17-A 18-B 19-B 20-B 21-A 22-D 23-C 24-A